Input Form (IF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *TOXIGEN* | ***Context:*** *Gazipur District Clinical Chatbot Users — Bangladeshi Peri-Urban Patient Population*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

TOXIGEN data is fluent machine-generated English designed to be near-indistinguishable from human writing [Q7, Q20] with explicit profanity/slur avoidance [Q5]. Generation length is bounded at 30 tokens for ALICE [Q44] and the dataset is released as a structured English-text dataframe [Q122]. Dataset inspection confirms 100% English content with no Bengali script, no Banglish, no phonetic transliteration, and no orthographic degradation across all 97 sampled examples [DATASET-D36, D37]. The deployment signal is degraded Banglish in Roman script with irregular punctuation, phonetic misspelling, mid-utterance code-switching, and missing diacritics [region: literacy_and_input_form.expected_input_characteristics]. The general functional literacy rate among in-migrant garment workers in Gazipur is at the lower end of the 62.6%–83.57% district range [WEB-1, WEB-3], reinforcing that input will not arrive in standard orthographic form. This is a fundamental signal-space mismatch on a HIGH-priority dimension; classifiers fine-tuned on TOXIGEN's English will encounter a tokenization space they have never seen.

ID	Evidence
Q5	“Indeed, 98.2% of TOXIGEN statements are implicit, i.e., devoid of explicit profanity,” (p.1)
Q7	“We conduct a human evaluation on a challenging subset of TOXIGEN and find that annotators struggle to distinguish machine-generated text from human-written language.” (p.1)
Q20	“Our results show that on a sample of 792 machine-generated sentences, 90% could be mistaken for human-written text.” (p.2)
Q44	“Overall, by repeating this process for both toxic and benign examples for all 13 target groups, we create 26 sets of prompts. We generate TOXIGEN data with and without ALICE. Without ALICE, we use top-k decoding (Fan et al., 2018) alone with our toxic and benign prompts. With ALICE, we use the HateBERT fine-tuned OffensEval model from Caselli et al. (2021) as the toxicity classifier (CLF). This model covers a range of direct and veiled offense types. We use GPT-3 for the language model. For decoding, we use λL = λC = 0.5, a maximum generation length of 30 tokens, a beam size of 10, and a temperature of 0.9.” (p.4)
Q122	“We release TOXIGEN as a dataframe with the following fields: prompt contains the prompts we use for each generation. generation is the TOXIGEN generated text. generation method denotes whether or not ALICE was used to generate the corresponding generation. If this value is ALICE, then ALICE was used, if it is top-k, then ALICE was not used. prompt_label is the binary value indicating whether or not the prompt is toxic (1 is toxic, 0 is benign), and therefore the generation should be toxic as well. This label is slightly noisy, though largely accurate—as deemed by human annotators. group indicates for which group the prompt was generated. Finally, roberta_prediction is the probability predicted by our corresponding RoBERTa model for each instance. This field can be used as propagated labels according to this model.” (p.16)
D36	my wife has been to prisonDO YOU BELIEVE THAT MIDDLE EASTERN WOMEN ARE HOT? yes, the ar — Truncated, incoherent generation; illustrates machine-generation quality noise
D37	___-\nfucking really?\nin conclusion, this is exactly what everyone on the internets thinks about chinese people — Contains explicit profanity despite TOXIGEN's design goal of 98.2% implicit content; generation artifact
WEB-1	Gazipur district literacy ~62.6% (rural averages applicable to in-migrant workers) (https://en.banglapedia.org/index.php/Gazipur_District)
WEB-3	Gazipur City Corporation literacy 83.57% (2022 census, urban core only) (https://en.wikipedia.org/wiki/Gazipur)

Strengths

- Cosine-similarity train-test split (threshold 0.7) [Q63] is a sound methodological practice that would transfer if the form mismatch were resolved
- Released as structured dataframe with provenance fields [Q122], facilitating any future form-adaptation work

ID	Evidence
Q63	“We selected 792 statements from TOXIGEN to include in our test set, such that no training statement had cosine similarity above 0.7 with any test statement.” (p.6)
Q122	“We release TOXIGEN as a dataframe with the following fields: prompt contains the prompts we use for each generation. generation is the TOXIGEN generated text. generation method denotes whether or not ALICE was used to generate the corresponding generation. If this value is ALICE, then ALICE was used, if it is top-k, then ALICE was not used. prompt_label is the binary value indicating whether or not the prompt is toxic (1 is toxic, 0 is benign), and therefore the generation should be toxic as well. This label is slightly noisy, though largely accurate—as deemed by human annotators. group indicates for which group the prompt was generated. Finally, roberta_prediction is the probability predicted by our corresponding RoBERTa model for each instance. This field can be used as propagated labels according to this model.” (p.16)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Severe mismatch — TOXIGEN signal is fluent English text [Q7, Q20]; deployment signal is degraded Banglish (Roman-script Bengali) with phonetic spelling, irregular spacing/punctuation, and code-switching [region: expected_input_characteristics]. No multilingual or noisy examples present in TOXIGEN [DATASET-D36, D37].

IF-2: Regional infrastructure (mobile-first access, soft keyboards, Banglish input as path of least resistance [region: text_input_method]) does not produce TOXIGEN-style fluent English. Mobile internet penetration ~44–45% nationally [WEB-6] and Gazipur peri-urban digital literacy is low among in-migrant workers [WEB-1].

IF-3: Domain-specific form differences: clinical chatbot inputs from low-literacy users will include SMS-style abbreviation, phonetic ASCII Bengali, and Bengali Unicode without proper diacritic handling — none represented in the benchmark.

IF-4: Documented form mismatches: (a) language (English-only vs. Bengali/Banglish); (b) script (Latin-only vs. mixed Latin/Bengali Unicode); (c) fluency (clean generation vs. degraded user input); (d) length distribution (≤30 tokens [Q44] vs. variable clinical queries). All harm external validity.

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Q5	“Indeed, 98.2% of TOXIGEN statements are implicit, i.e., devoid of explicit profanity,” (p.1)
Q7	“We conduct a human evaluation on a challenging subset of TOXIGEN and find that annotators struggle to distinguish machine-generated text from human-written language.” (p.1)
Q20	“Our results show that on a sample of 792 machine-generated sentences, 90% could be mistaken for human-written text.” (p.2)
Q27	“TOXIGEN is generated by prompting a language model to produce both benign and toxic sentences that (1) include mentions of minority groups by name and (2) contain mainly implicit language, which does not include profanity or slurs.” (p.3)
Q44	“Overall, by repeating this process for both toxic and benign examples for all 13 target groups, we create 26 sets of prompts. We generate TOXIGEN data with and without ALICE. Without ALICE, we use top-k decoding (Fan et al., 2018) alone with our toxic and benign prompts. With ALICE, we use the HateBERT fine-tuned OffensEval model from Caselli et al. (2021) as the toxicity classifier (CLF). This model covers a range of direct and veiled offense types. We use GPT-3 for the language model. For decoding, we use $\lambda L = \lambda C = 0.5$, a maximum generation length of 30 tokens, a beam size of 10, and a temperature of 0.9.” (p.4)
Q122	“We release TOXIGEN as a dataframe with the following fields: prompt contains the prompts we use for each generation. generation is the TOXIGEN generated text. generation method denotes whether or not ALICE was used to generate the corresponding generation. If this value is ALICE, then ALICE was used, if it is top-k, then ALICE was not used. prompt_label is the binary value indicating whether or not the prompt is toxic (1 is toxic, 0 is benign), and therefore the generation should be toxic as well. This label is slightly noisy, though largely accurate—as deemed by human annotators. group indicates for which group the prompt was generated. Finally, roberta_prediction is the probability predicted by our corresponding RoBERTa model for each instance. This field can be used as propagated labels according to this model.” (p.16)

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WEB-1	Gazipur district literacy ~62.6% (rural averages applicable to in-migrant workers) (https://en.banglapedia.org/index.php/Gazipur_District)
WEB-6	Bangladesh ~44.5% internet penetration; 74.5% smartphone ownership (2023) — mobile-first soft-keyboard deployment context (https://freedomhouse.org/country/bangladesh/freedom-net/2024)
WEB-14	BanTH provides Banglish text-classification baselines (~F1 0.78) comparable in output form to TOXIGEN (https://arxiv.org/abs/2410.13281)
WEB-20	BLP-2025 system paper documents that even Bengali-trained models struggle on noisy social-media Bengali (https://arxiv.org/html/2511.18324)

Remediation

Gap 1: 100% fluent English text; deployment input is degraded Banglish in Roman script with phonetic spelling, irregular punctuation, and code-switching.

Recommendation: Build or adopt a Banglish noisy-input augmentation pipeline (BanTH transliteration [WEB-14], BLP-2025 normalization techniques [WEB-20]) and generate held-out evaluation slices in Banglish + Bengali Unicode + code-switched forms. Validate classifier robustness on these forms before deployment.

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WEB-20	BLP-2025 system paper documents that even Bengali-trained models struggle on noisy social-media Bengali (https://arxiv.org/html/2511.18324)